7. General Curvilinear Coordinates

7.30. Curvilinear coordinate systems

For each point P with Cartesian coordinates (x, y, z), associate a unique set of curvilinear coordinates (u_1, u_2, u_3) where

$$\begin{aligned} x =& x(u_1, u_2, u_3) \\ y =& y(u_1, u_2, u_3) \\ z =& z(u_1, u_2, u_3). \end{aligned}$$

and

$$\begin{aligned} u_1 =& u_1(x, y, z) \\ u_2 =& u_2(x, y, z) \\ u_3 =& u_3(x, y, z). \end{aligned}$$

Examples: cylindrical coordinates; spherical coordinates

Unit Vectors

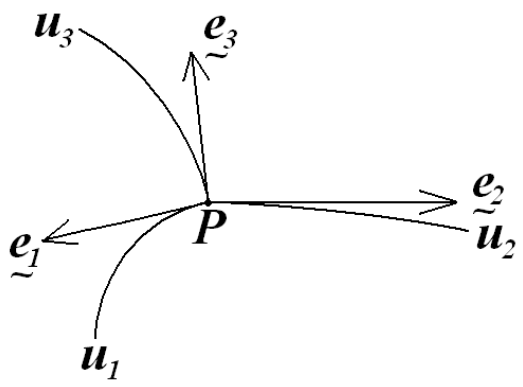
Let (x, y, z) be the position vector of P. We can express it as

$$\mathbf{r}(u_1, u_2, u_3) = (x(u_1, u_2, u_3), y(u_1, u_2, u_3), z(u_1, u_2, u_3))$$

where we set $\mathbf{r} = (x, y, z)$.

- If we hold $u_2,\ u_3$ constant and let u_1 vary, then we get a curve passing through P and a tangent vector at P is $\dfrac{\partial \mathbf{r}}{\partial u_1}$.
  The unit tangent vector in the direction of u_1 is

$$\mathbf{e}_1 = \frac{\dfrac{\partial \mathbf{r}}{\partial u_1}}{\left|\dfrac{\partial \mathbf{r}}{\partial u_1}\right|} \Rightarrow \frac{\partial \mathbf{r}}{\partial u_1} = \left|\frac{\partial \mathbf{r}}{\partial u_1}\right| \mathbf{e}_1 = h_1\, \mathbf{e}_1$$



- Similarly, tangent vectors at P in the direction of u_2 and u_3 are

$$\frac{\partial \mathbf{r}}{\partial u_2} = \left| \frac{\partial \mathbf{r}}{\partial u_2} \right| \mathbf{e}_2 = h_2\, \mathbf{e}_2$$
$$\frac{\partial \mathbf{r}}{\partial u_3} = \left| \frac{\partial \mathbf{r}}{\partial u_3} \right| \mathbf{e}_3 = h_3\, \mathbf{e}_3$$

Note

1. $h_1,\ h_2,\ h_3$ are called scale factors.
2. A curvilinear coordinate system is said to be *orthogonal* if

$$\mathbf{e}_i \cdot \mathbf{e}_j = 0, \text{ for } i \neq j.$$

Example 30.1

Cylindrical coordinates (ρ, ϕ, z):

$$x = \rho \cos \phi, \;\; y = \rho \sin \phi, \;\; z = z$$

$$0 \leq \phi \leq 2\pi, \;\; \rho \geq 0, \;\; z \in \mathbb{R}.$$

Find the scale factors and unit tangent vectors.

Example 30.2

Spherical Coordinates (r, θ, ϕ)

$$x = r\sin\theta\cos\phi,\ \ y = r\sin\theta\sin\phi,\ \ z = r\cos\theta$$

$$0 \le \phi \le 2\pi,\ \ 0 \le \theta \le \pi,\ \ r \ge 0$$

Find the scale factors and unit tangent vectors.

Tangents to Curves

Let u_1, u_2, u_3 be curvilinear coordinates. Consider the parametric curve

$$\mathbf{c}(t) = \mathbf{r}\left(u_1(t), u_2(t), u_3(t)\right).$$

Using the chain rule, the tangent to $\mathbf{c}(t)$ is

$$\begin{aligned} \frac{d\mathbf{c}}{dt} &= \frac{\partial \mathbf{r}}{\partial u_1}\frac{du_1}{dt} + \frac{\partial \mathbf{r}}{\partial u_2}\frac{du_2}{dt} + \frac{\partial \mathbf{r}}{\partial u_3}\frac{du_3}{dt} \\ \Rightarrow \frac{d\mathbf{c}}{dt} &= h_1\mathbf{e}_1\frac{du_1}{dt} + h_2\mathbf{e}_2\frac{du_2}{dt} + h_3\mathbf{e}_3\frac{du_3}{dt}. \end{aligned}$$

Example 30.3

Let

$$\mathbf{c}(t) = 2\cos(3t)\,\mathbf{i} + 2\sin(3t)\,\mathbf{j} + \mathbf{k}\,.$$

Write $\dfrac{d\mathbf{c}}{dt}$ in cylindrical coordinates.

7.31. Volume and Surface Area Elements

Volume Element

When we change variables from (x, y, z) to (u_1, u_2, u_3), the Jacobian is

$$\det \begin{bmatrix} \frac{\partial x}{\partial u_1} & \frac{\partial x}{\partial u_2} & \frac{\partial x}{\partial u_3} \\ \frac{\partial y}{\partial u_1} & \frac{\partial y}{\partial u_2} & \frac{\partial y}{\partial u_3} \\ \frac{\partial z}{\partial u_1} & \frac{\partial z}{\partial u_2} & \frac{\partial z}{\partial u_3} \end{bmatrix} = \det \begin{bmatrix} \frac{\partial x}{\partial u_1} & \frac{\partial y}{\partial u_1} & \frac{\partial z}{\partial u_1} \\ \frac{\partial x}{\partial u_2} & \frac{\partial y}{\partial u_2} & \frac{\partial z}{\partial u_2} \\ \frac{\partial x}{\partial u_3} & \frac{\partial y}{\partial u_3} & \frac{\partial z}{\partial u_3} \end{bmatrix}$$

$$= \left(\frac{\partial x}{\partial u_1}, \frac{\partial y}{\partial u_1}, \frac{\partial z}{\partial u_1} \right) \cdot \left[\left(\frac{\partial x}{\partial u_2}, \frac{\partial y}{\partial u_2}, \frac{\partial z}{\partial u_2} \right) \times \left(\frac{\partial x}{\partial u_3}, \frac{\partial y}{\partial u_3}, \frac{\partial z}{\partial u_3} \right) \right].$$

$$\Rightarrow \text{Jacobian} = \frac{\partial \mathbf{r}}{\partial u_1} \cdot \left(\frac{\partial \mathbf{r}}{\partial u_2} \times \frac{\partial \mathbf{r}}{\partial u_3} \right)$$
$$= (h_1\, \mathbf{e}_1) \cdot ((h_2\, \mathbf{e}_2) \times (h_3\, \mathbf{e}_3))$$
$$= h_1 h_2 h_3\ \mathbf{e}_1 \cdot (\mathbf{e}_2 \times \mathbf{e}_3)$$

$$\Rightarrow |\text{Jacobian}| = h_1 h_2 h_3\ \ |\mathbf{e}_1 \cdot (\mathbf{e}_2 \times \mathbf{e}_3)|$$

- If $\mathbf{e}_1,\ \mathbf{e}_2,\ \mathbf{e}_3$ are orthogonal unit vectors then $|\mathbf{e}_1 \cdot (\mathbf{e}_2 \times \mathbf{e}_3)| = 1$, so

$$|\text{Jacobian}| = h_1 h_2 h_3$$
$$dV = \ dx\ dy\ dz = h_1 h_2 h_3\ du_1\ du_2\ du_3$$

- cylindrical coordinates

$$|\text{Jacobian}| = h_\rho h_\phi h_z = \rho$$

- spherical coordinates

$$|\text{Jacobian}| = h_r h_\theta h_\phi = r^2 \sin\theta$$

Surface Area Element

If S is a surface parametrised by

$$x = f(u, v),\ y = g(u, v),\ z = h(u, v)$$

then

$$dS = |\mathbf{T_u} \times \mathbf{T_v}|\ du\, dv.$$

We have

$$\begin{aligned} |\mathbf{T_u} \times \mathbf{T_v}| &= \left| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right| \\ &= |(h_u \mathbf{e}_u) \times (h_v \mathbf{e}_v)| \\ &= h_u h_v\, |\mathbf{e}_u \times \mathbf{e}_v|\,. \end{aligned}$$

If $\mathbf{e}_u,\ \mathbf{e}_v$ are orthogonal unit vectors then $|\mathbf{e}_u \times \mathbf{e}_v| = 1$. Thus

$$|\mathbf{T_u} \times \mathbf{T_v}| = h_u h_v \implies dS = h_u h_v\, du\, dv.$$

Example 31.1

Evaluate $\iint\limits_S z\, dS$ where S is the cylinder $x^2 + y^2 = 9,\ 0 \le z \le 5$.

Example 31.2

Evaluate $\iint\limits_S z\, dS$ where S is the hemisphere $x^2 + y^2 + z^2 = 4,\ z \geq 0$.

7.32. Grad, Div, Curl, and Laplacian in Orthogonal Curvilinear Coordinates

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a C^2 scalar function and let $\mathbf{F} : \mathbb{R}^3 \to \mathbb{R}^3$ be a C^1 vector field where

$$\mathbf{F}(u_1, u_2, u_3) = F_1\mathbf{e}_1 + F_2\mathbf{e}_2 + F_3\mathbf{e}_3.$$

Here $\mathbf{e}_1$, $\mathbf{e}_2$ and $\mathbf{e}_3$ are *orthogonal* curvilinear coordinates. Then by the chain rule and orthogonality we have

1. $\boldsymbol{\nabla} f = \dfrac{1}{h_1}\dfrac{\partial f}{\partial u_1}\mathbf{e}_1 + \dfrac{1}{h_2}\dfrac{\partial f}{\partial u_2}\mathbf{e}_2 + \dfrac{1}{h_3}\dfrac{\partial f}{\partial u_3}\mathbf{e}_3$
2. $\boldsymbol{\nabla} \cdot \mathbf{F} = \dfrac{1}{h_1 h_2 h_3}\left[\dfrac{\partial\left(h_2 h_3 F_1\right)}{\partial u_1} + \dfrac{\partial\left(h_1 h_3 F_2\right)}{\partial u_2} + \dfrac{\partial\left(h_1 h_2 F_3\right)}{\partial u_3}\right]$

3. $\boldsymbol{\nabla} \times \mathbf{F} = \dfrac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1\mathbf{e}_1 & h_2\mathbf{e}_2 & h_3\mathbf{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 F_1 & h_2 F_2 & h_3 F_3 \end{vmatrix}$

4.

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial u_1}\left(\frac{h_2 h_3}{h_1}\frac{\partial f}{\partial u_1}\right) + \frac{\partial}{\partial u_2}\left(\frac{h_1 h_3}{h_2}\frac{\partial f}{\partial u_2}\right) + \frac{\partial}{\partial u_3}\left(\frac{h_1 h_2}{h_3}\frac{\partial f}{\partial u_3}\right)\right].$$

Note

Equations 1-4 reduce to the usual expressions for cartesian coordinates if

$$h_1 = h_2 = h_3 = 1, \quad (u_1, u_2, u_3) = (x, y, z),$$
$$\mathbf{e}_1 = \mathbf{i}\,,\ \mathbf{e}_2 = \mathbf{j}\,,\ \mathbf{e}_3 = \mathbf{k}\,.$$

Cylindrical Coordinates

Cylindrical coordinates (ρ, ϕ, z) are defined by

$$x = \rho \cos\phi, \quad y = \rho \sin\phi, \quad z = z$$

where $\rho \geq 0$, $0 \leq \phi \leq 2\pi$.

Then $(u_1, u_2, u_3) = (\rho, \phi, z)$ and $h_1 = 1$, $h_2 = \rho$, $h_3 = 1$.

Equations 1-4 reduce to

1. $\boldsymbol{\nabla} f = \dfrac{\partial f}{\partial \rho}\,\hat{\boldsymbol{\rho}} + \dfrac{1}{\rho}\,\dfrac{\partial f}{\partial \phi}\,\hat{\boldsymbol{\phi}} + \dfrac{\partial f}{\partial z}\,\hat{\boldsymbol{z}}$

2. $$\boldsymbol{\nabla} \cdot \mathbf{F} = \frac{1}{\rho}\left[\frac{\partial(\rho F_1)}{\partial \rho} + \frac{\partial F_2}{\partial \phi} + \frac{\partial(\rho F_3)}{\partial z}\right] = \frac{1}{\rho}\left(F_1 + \rho\,\frac{\partial F_1}{\partial \rho}\right) + \frac{1}{\rho}\,\frac{\partial F_2}{\partial \phi} + \frac{\partial F_3}{\partial z}$$

Cylindrical Coordinates

3. $$\boldsymbol{\nabla} \times \mathbf{F} = \frac{1}{\rho} \begin{vmatrix} \hat{\boldsymbol{\rho}} & \rho\hat{\boldsymbol{\phi}} & \hat{\boldsymbol{z}} \\ \dfrac{\partial}{\partial \rho} & \dfrac{\partial}{\partial \phi} & \dfrac{\partial}{\partial z} \\ F_1 & \rho F_2 & F_3 \end{vmatrix}$$

4. $$\begin{aligned} \nabla^2 f &= \frac{1}{\rho}\left[\frac{\partial}{\partial \rho}\left(\rho\,\frac{\partial f}{\partial \rho}\right) + \frac{\partial}{\partial \phi}\left(\frac{1}{\rho}\,\frac{\partial f}{\partial \phi}\right) + \frac{\partial}{\partial z}\left(\rho\,\frac{\partial f}{\partial z}\right)\right] \\ &= \frac{1}{\rho}\,\frac{\partial f}{\partial \rho} + \frac{\partial^2 f}{\partial \rho^2} + \frac{1}{\rho^2}\,\frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2} \end{aligned}$$

Spherical Coordinates

Spherical coordinates (r, θ, ϕ) are defined by

$$x = r\sin\theta\cos\phi, \quad y = r\sin\theta\sin\phi, \quad z = r\cos\theta$$

where $r \geq 0$, $0 \leq \theta \leq \pi$, $0 \leq \phi \leq 2\pi$.

Then $(u_1, u_2, u_3) = (r, \theta, \phi)$ and $h_1 = 1$, $h_2 = r$, $h_3 = r\sin\theta$.

Equations 1-4 reduce to

1. $\boldsymbol{\nabla} f = \dfrac{\partial f}{\partial r}\,\hat{\boldsymbol{r}} + \dfrac{1}{r}\,\dfrac{\partial f}{\partial \theta}\,\hat{\boldsymbol{\theta}} + \dfrac{1}{r\sin\theta}\,\dfrac{\partial f}{\partial \phi}\,\hat{\boldsymbol{\phi}}$
2. $\boldsymbol{\nabla}\cdot\mathbf{F} = \dfrac{1}{r^2\sin\theta}\left[\dfrac{\partial\left(r^2\sin\theta F_1\right)}{\partial r} + \dfrac{\partial\left(r\sin\theta F_2\right)}{\partial\theta} + \dfrac{\partial\left(rF_3\right)}{\partial\phi}\right]$

$$= \frac{1}{r^2}\,\frac{\partial\left(r^2F_1\right)}{\partial r} + \frac{1}{r\sin\theta}\,\frac{\partial\left(\sin\theta F_2\right)}{\partial\theta} + \frac{1}{r\sin\theta}\,\frac{\partial F_3}{\partial\phi}$$

Spherical Coordinates

3. $$\boldsymbol{\nabla} \times \mathbf{F} = \frac{1}{r^2 \sin\theta} \begin{vmatrix} \hat{\boldsymbol{r}} & r\hat{\boldsymbol{\theta}} & r\sin\theta\hat{\boldsymbol{\phi}} \\ \dfrac{\partial}{\partial r} & \dfrac{\partial}{\partial \theta} & \dfrac{\partial}{\partial \phi} \\ F_1 & rF_2 & r\sin\theta F_3 \end{vmatrix}$$

4. $$\nabla^2 f = \frac{1}{r^2 \sin\theta}\left[\frac{\partial}{\partial r}\left(r^2 \sin\theta \frac{\partial f}{\partial r}\right) + \frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial f}{\partial \theta}\right) + \frac{\partial}{\partial \phi}\left(\frac{1}{\sin\theta}\frac{\partial f}{\partial \phi}\right)\right]$$

$$= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial f}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 f}{\partial \phi^2}$$

Example 32.1

Let $f(r,\theta,\phi) = r\theta\phi, \mathbf{F}(r,\theta,\phi) = \sin\theta\,\hat{\mathbf{r}} + r\,\hat{\phi}$.
Find $\boldsymbol{\nabla} f,\ \boldsymbol{\nabla}\cdot\mathbf{F},\ \boldsymbol{\nabla}\times\mathbf{F},\ \nabla^2 f$ in spherical coordinates.

Example 32.2

Express the heat equation

$$\frac{\partial u}{\partial t} = k\nabla^2 u$$

in spherical coordinates if u depends only on r and t.

Example 32.3

Redo Example (15.5) of basic identities of vector calculus.

Find $\boldsymbol{\nabla} \cdot \left(\frac{\mathbf{r}}{r^2} \right)$ where

$\mathbf{r}(x, y, z) = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}\,,\ r = \sqrt{x^2 + y^2 + z^2}$.

Example 32.4

Redo Example (15.7) of basic identities of vector calculus.
Find $\nabla^2(r^2 \log(r))$ where
$\mathbf{r}(x, y, z) = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}\,,\ r = \sqrt{x^2 + y^2 + z^2}$.